

Lab09

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IMPORTING CANDY DATA

```
candy_file <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-rankings.csv"
candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

85

Q2. How many fruity candy types are in the dataset?

38

```
#Q1. Number of candy types  
dim(candy)
```

```
[1] 85 12
```

```
#Q2. Number of fruity candy  
sum(candy$fruity == 1)
```

```
[1] 38
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

Milky Way, winpercent = 73.09956

Q4. What is the winpercent value for "Kit Kat"?

76.7686

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

49.6535

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
#Q3. winpercent of favorite candy, "Milky Way"  
candy |>  
  filter(row.names(candy)=="Milky Way") |>  
  select(winpercent)
```

```
      winpercent
Milky Way    73.09956
```

```
#Q4. winpercent of "Kit Kat"
candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat    76.7686
```

```
#Q5. Winpercent of "Tootsie Roll Snack Bars"
candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
      winpercent
Tootsie Roll Snack Bars    49.6535
```

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes. Most variables are on the 0-1 scale, but “winpercent” stands out because it is a continuous numeric variable.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

In the chocolate column, 0 indicates the candy does not contain chocolate, and the 1 indicates that the candy does contain chocolate.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12

Column type frequency:

numeric

12

Group variables

None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

```
#Q6 code
summary(candy)
```

chocolate	fruity	caramel	peanutyalmondy
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.4353	Mean :0.4471	Mean :0.1647	Mean :0.1647
3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :1.0000	Max. :1.0000	Max. :1.0000	Max. :1.0000
nougat	crispedricewafer	hard	bar
Min. :0.00000	Min. :0.00000	Min. :0.0000	Min. :0.0000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.00000	Median :0.00000	Median :0.0000	Median :0.0000
Mean :0.08235	Mean :0.08235	Mean :0.1765	Mean :0.2471
3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.0000	3rd Qu.:0.0000
Max. :1.00000	Max. :1.00000	Max. :1.0000	Max. :1.0000

pluribus	sugarpercent	pricepercent	winpercent
Min. :0.0000	Min. :0.0110	Min. :0.0110	Min. :22.45
1st Qu.:0.0000	1st Qu.:0.2200	1st Qu.:0.2550	1st Qu.:39.14
Median :1.0000	Median :0.4650	Median :0.4650	Median :47.83
Mean :0.5176	Mean :0.4786	Mean :0.4689	Mean :50.32
3rd Qu.:1.0000	3rd Qu.:0.7320	3rd Qu.:0.6510	3rd Qu.:59.86
Max. :1.0000	Max. :0.9880	Max. :0.9760	Max. :84.18

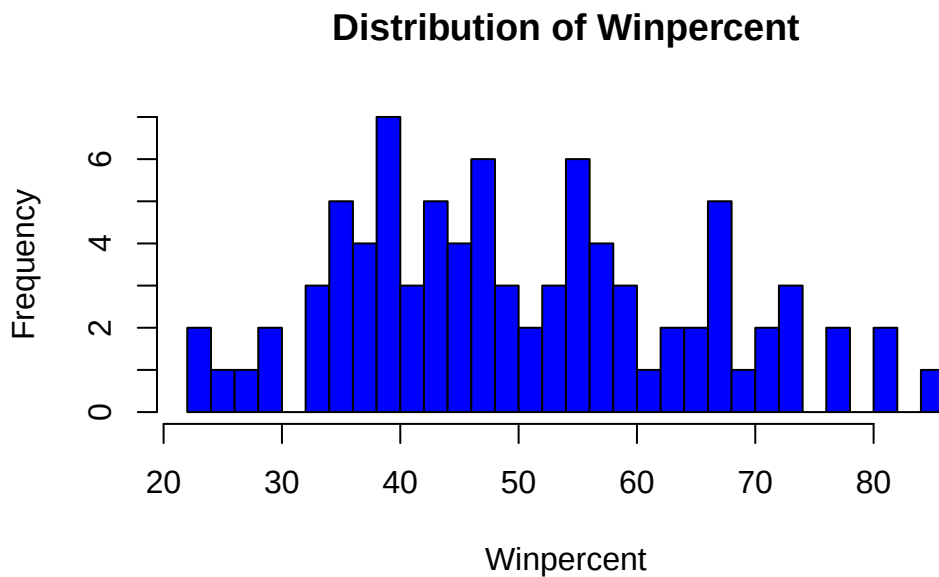
```
#Q7 code
table(candy$chocolate)
```

```
0 1
48 37
```

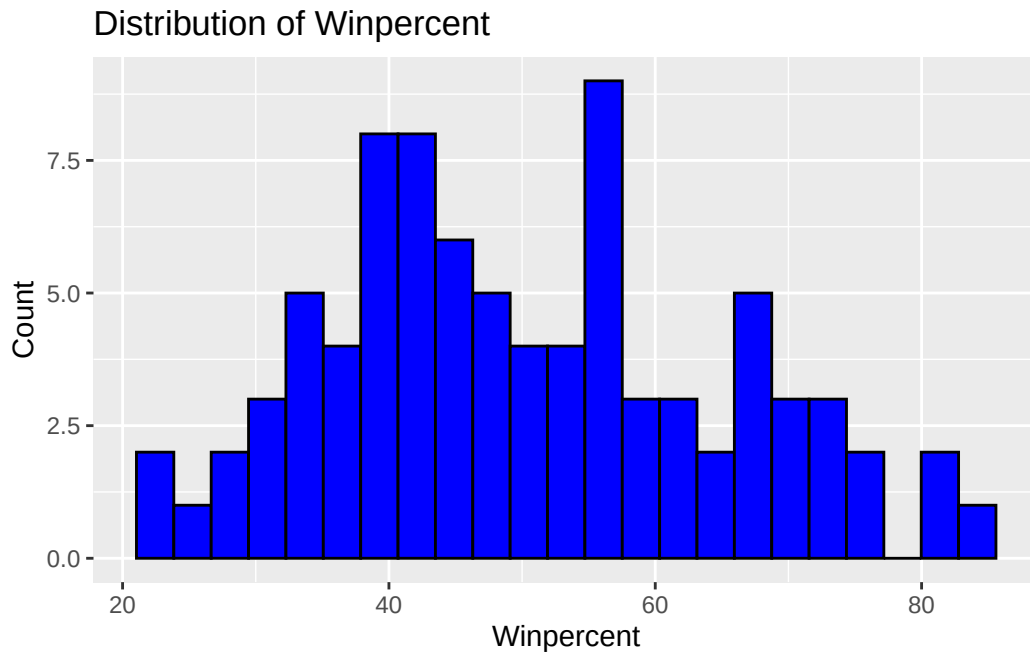
EXPLORATORY ANALYSIS

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
#Q8. Histogram of winpercent...
##Method 1: Using Base R
hist(candy$winpercent, main = "Distribution of Winpercent",
     xlab = "Winpercent", col = "blue", breaks = 23)
```



```
##Method 2: Using ggplot2
library(ggplot2)
ggplot(candy, aes(x=winpercent)) +
  geom_histogram(bins = 23, fill = "blue", color = "black") +
  labs(title = "Distribution of Winpercent",
       x = "Winpercent", y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical?

No, it seems kind of skewed.

Q10. Is the center of the distribution above or below 50%?

The center of distribution is above 50%, as the mean (50.3%) is above 50%.

```
#Q10 Center of distribution above or below?
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
median(candy$winpercent)
```

```
[1] 47.82975
```

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

On average, chocolate candies are ranked higher than fruit candies.

```
#Q11 Code
```

```
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

Since the p-value is 2.871e-08, which is < 0.05 , the difference is statistically significant.

```
#Q12 t.test
```

```
t.test(candy$winpercent[as.logical(candy$chocolate)],  
       candy$winpercent[as.logical(candy$fruity)])
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

OVERALL CANDY RANKINGS

Q13. What are the five least liked candy types in this set?

Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

#Q13 Code using base R

```
head(candy[order(candy$winpercent),], n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat		
Nik L Nip	0	1	0			0	0	
Boston Baked Beans	0	0	0			1	0	
Chiclets	0	1	0			0	0	
Super Bubble	0	1	0			0	0	
Jawbusters	0	1	0			0	0	

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197	0.976	
Boston Baked Beans				0	0	0	1	0.313	0.511	
Chiclets				0	0	0	1	0.046	0.325	
Super Bubble				0	0	0	0	0.162	0.116	
Jawbusters				0	1	0	1	0.093	0.511	

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

#Q13 Code using dplyr

```
candy |> arrange(winpercent) |> head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat		
Nik L Nip	0	1	0			0	0	
Boston Baked Beans	0	0	0			1	0	
Chiclets	0	1	0			0	0	
Super Bubble	0	1	0			0	0	
Jawbusters	0	1	0			0	0	

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197	0.976	
Boston Baked Beans				0	0	0	1	0.313	0.511	
Chiclets				0	0	0	1	0.046	0.325	
Super Bubble				0	0	0	0	0.162	0.116	

Jawbusters		0	1	0	1	0.093	0.511
	winpercent						
Nik L Nip	22.44534						
Boston Baked Beans	23.41782						
Chiclets	24.52499						
Super Bubble	27.30386						
Jawbusters	28.12744						

Q14. What are the top 5 all time favorite candy types out of this set?

Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

#Q14 Code using base R

```
head(candy[order(candy$winpercent, decreasing = TRUE),], n = 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar
Reese's Peanut Butter cup		0	0	0		0	0.720
Reese's Miniatures		0	0	0		0	0.034
Twix		1	0	1		0	0.546
Kit Kat		1	0	1		0	0.313
Snickers		0	0	1		0	0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651	84.18029	
Reese's Miniatures	0.279	81.86626	
Twix	0.906	81.64291	
Kit Kat	0.511	76.76860	
Snickers	0.651	76.67378	

#Q14 Code using dplyr

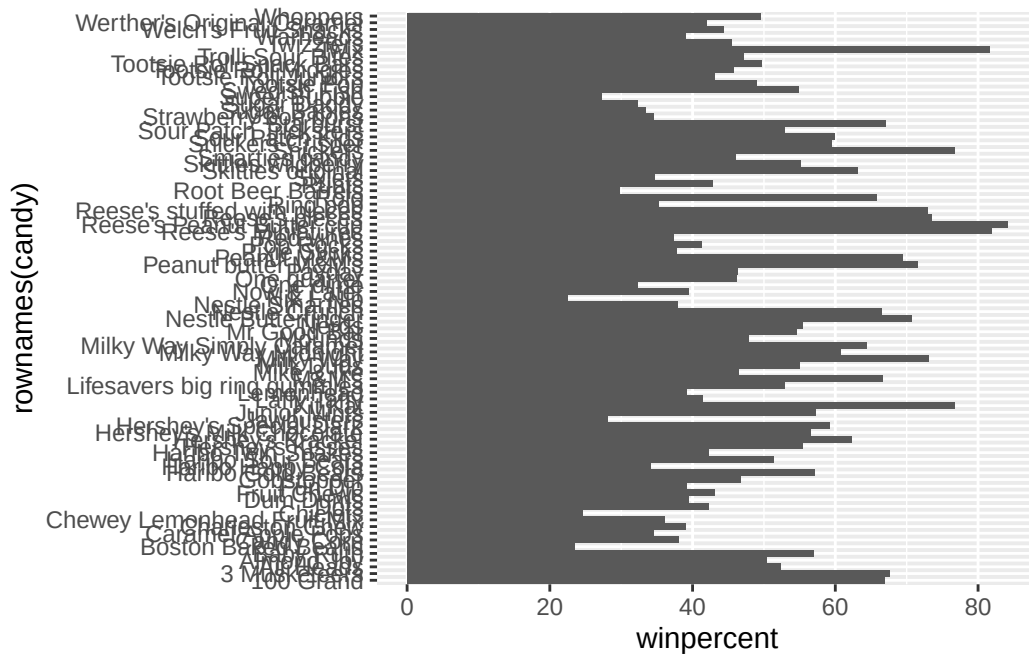
```
candy |> arrange(desc(winpercent)) |> head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0

Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1
	crisped	ricewafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup	0	0	0	0	0.720
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546
	pricepercent	winpercent			
Reese's Peanut Butter cup	0.651	84.18029			
Reese's Miniatures	0.279	81.86626			
Twix	0.906	81.64291			
Kit Kat	0.511	76.76860			
Snickers	0.651	76.67378			

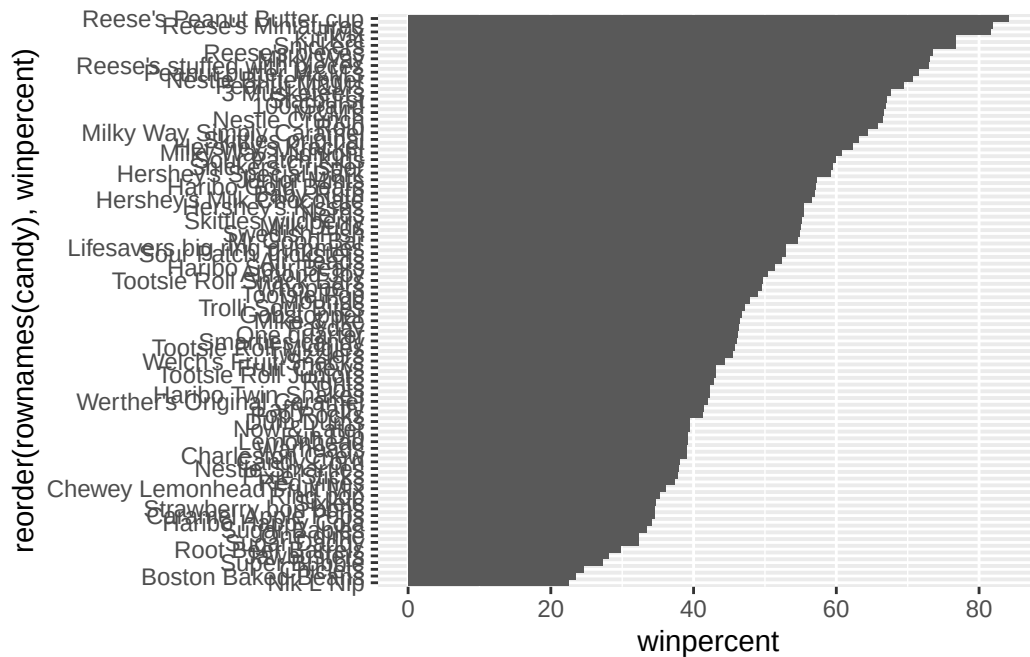
Q15. Make a first barplot of candy ranking based on winpercent values.

```
#Q15 barplot
ggplot(candy, aes(x = winpercent, y = rownames(candy))) +
  geom_col()
```



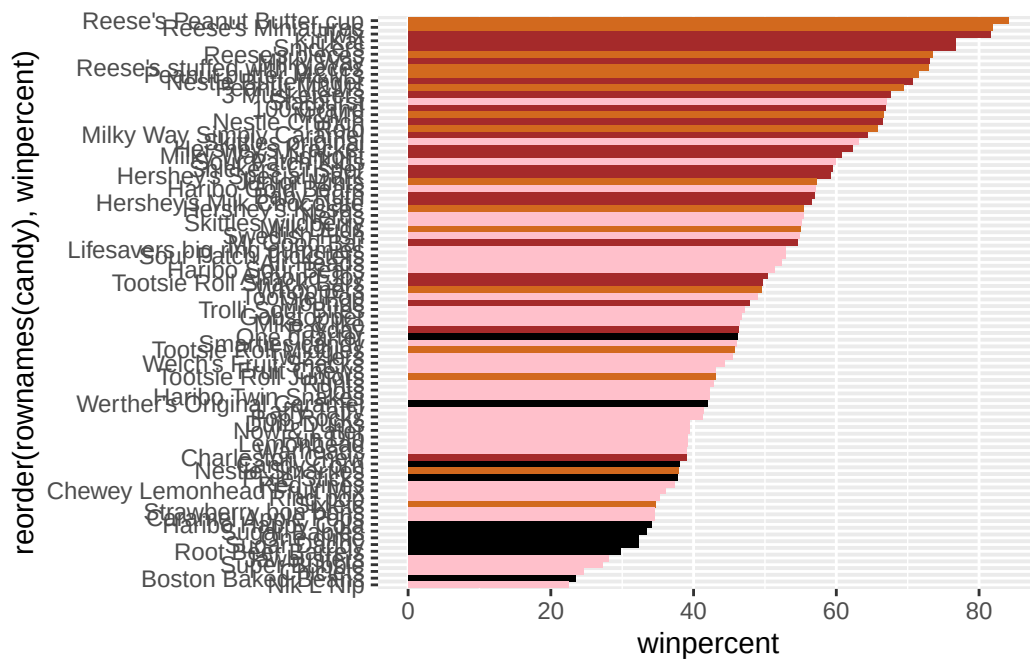
Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent.

```
#Q16 recorder bars to sort by winpercent
ggplot(candy, aes(winpercent, reorder(rownames(candy), winpercent))) +
  geom_col()
```



```
#Add color
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is “Sixlets”.

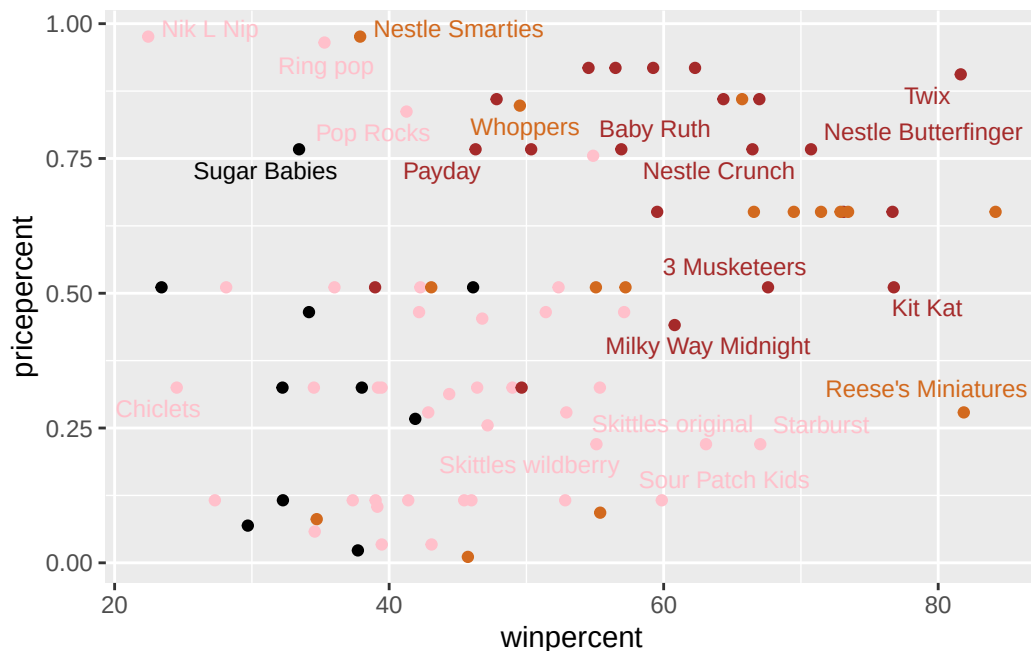
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is “Starburst”.

TAKING A LOOK AT PRICEPERCENT

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Peanut Butter Cup

```
#Q19 Ranking in terms of winpercent for the least money
candy$value_score <- candy$winpercent - candy$pricepercent
candy |> arrange(desc(value_score)) |> head(1)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
	crisp	edric	wafer	hard bar	pluribus	sugarpercent
Reese's Peanut Butter cup		0	0	0	0	0.72
	pricepercent	winpercent	value_score			
Reese's Peanut Butter cup	0.651	84.18029	83.52929			

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

Top 5 most expensive candy types: Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular among these is Nik L Nip.

```
#Q20 Price of Candy
ord <- order(candy$pricepercent, decreasing = TRUE)
head(candy[ord,c(11,12)], n=5)
```

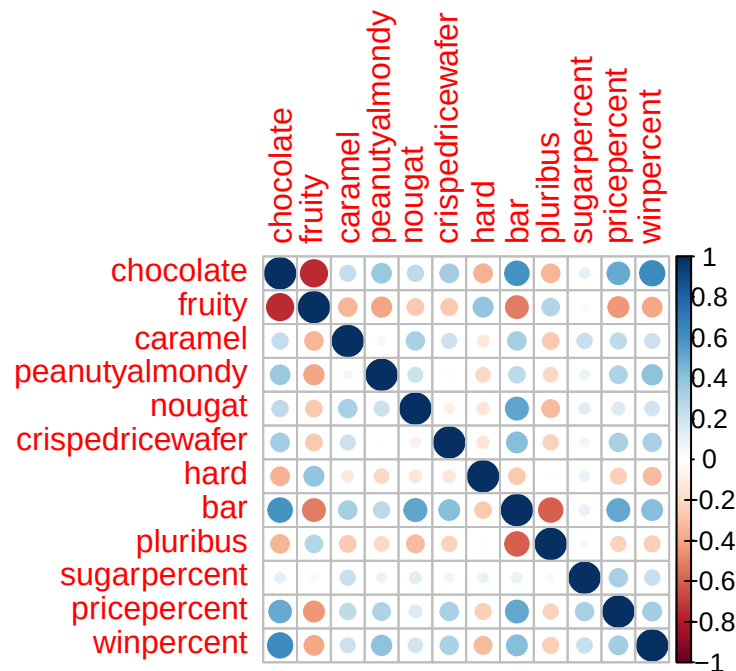
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

EXPLORING THE CORRELATION STRUCTURE

```
library(corrplot)
```

corrplot 0.95 loaded

```
candy_BK <- candy
candy <- candy %>% dplyr::select(-value_score)
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot, what two variables are anti-correlated (i.e. have minus values)?

There is a negative correlation between “fruity” and “chocolate”.

Q23. Similarly, what two variables are most positively correlated?

There is a positive correlation between “chocolate” and “bar”.

PRINCIPAL COMPONENT ANALYSIS

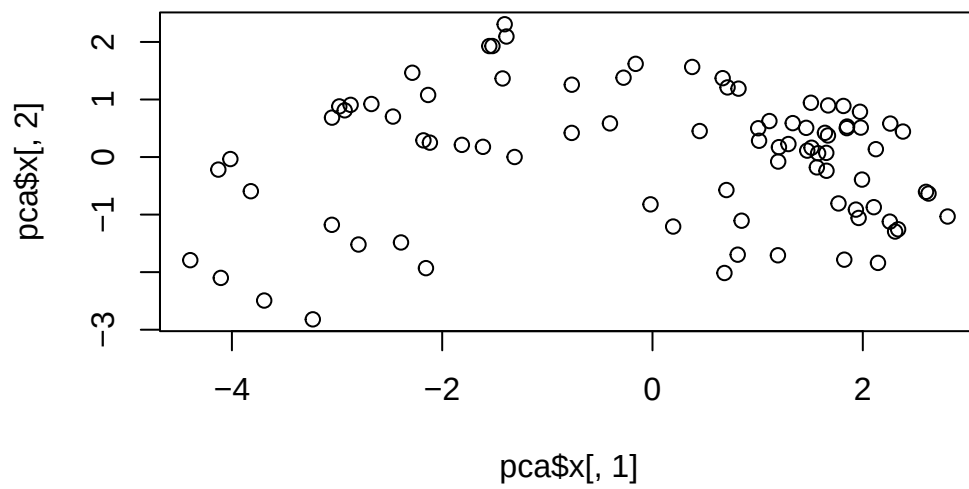
```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

Importance of components:

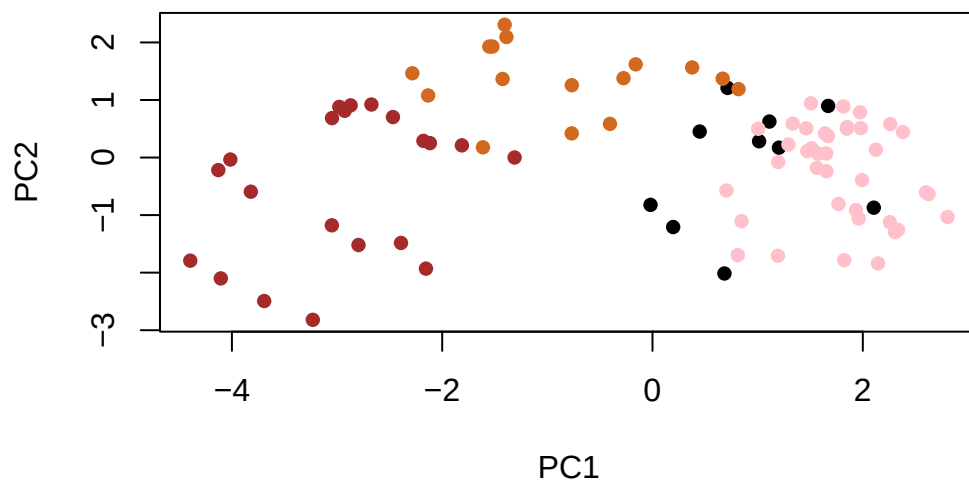
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1], pca$x[,2])
```



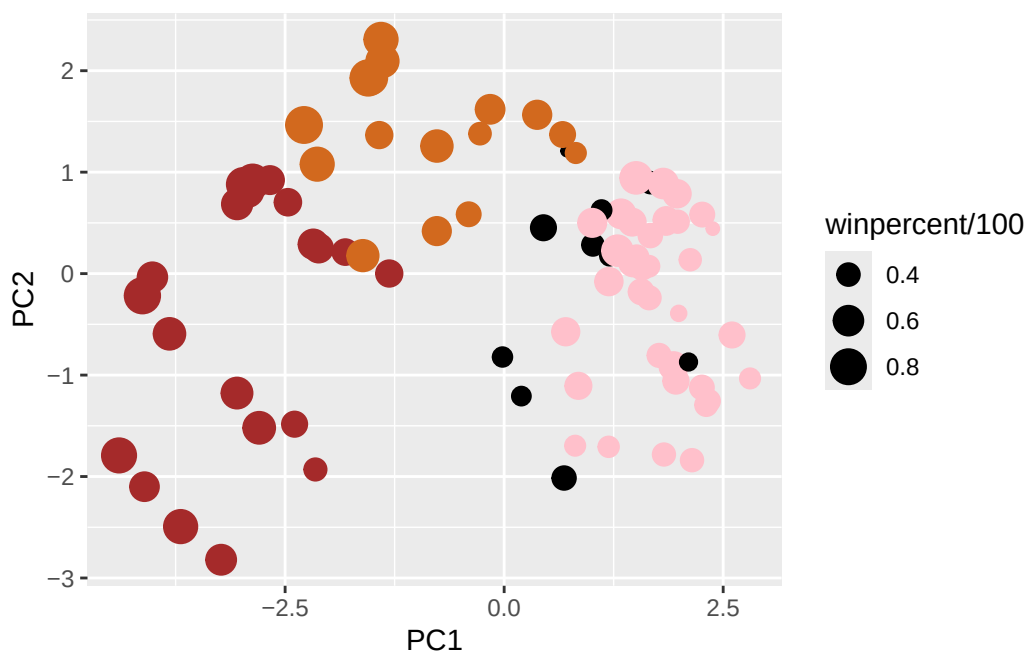
```
plot(pca$x[,1:2], col=my_cols, pch=16)
```




```
#Nicer plot with ggplot2
##Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col = my_cols)
```

p



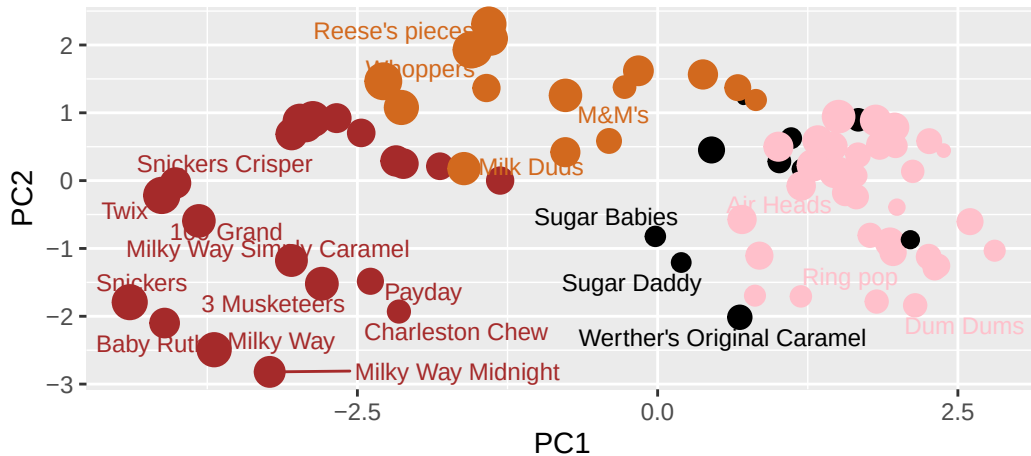
```
#Nicer plot with ggrepel
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown),
       chocolate other (light brown), fruity (red), other
       (black)",
```

```
caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown),
chocolate other (light brown), fruity (red), other
(black)



Data from 538

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Original variables that are picked up strongly by PC1 in the positive direction: fruity and pluribus (and hard). Yes, this does make sense because it represents a contrast between the two candy types: chocolate and fruity. We've seen this relationship in previous questions: in Q22, the "chocolate" and "fruity" variables were negatively correlated, and again in Q11/12, chocolate candies were ranked higher than fruity candies, which suggests the two types perform differently from each other.

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

```
last_plot
```

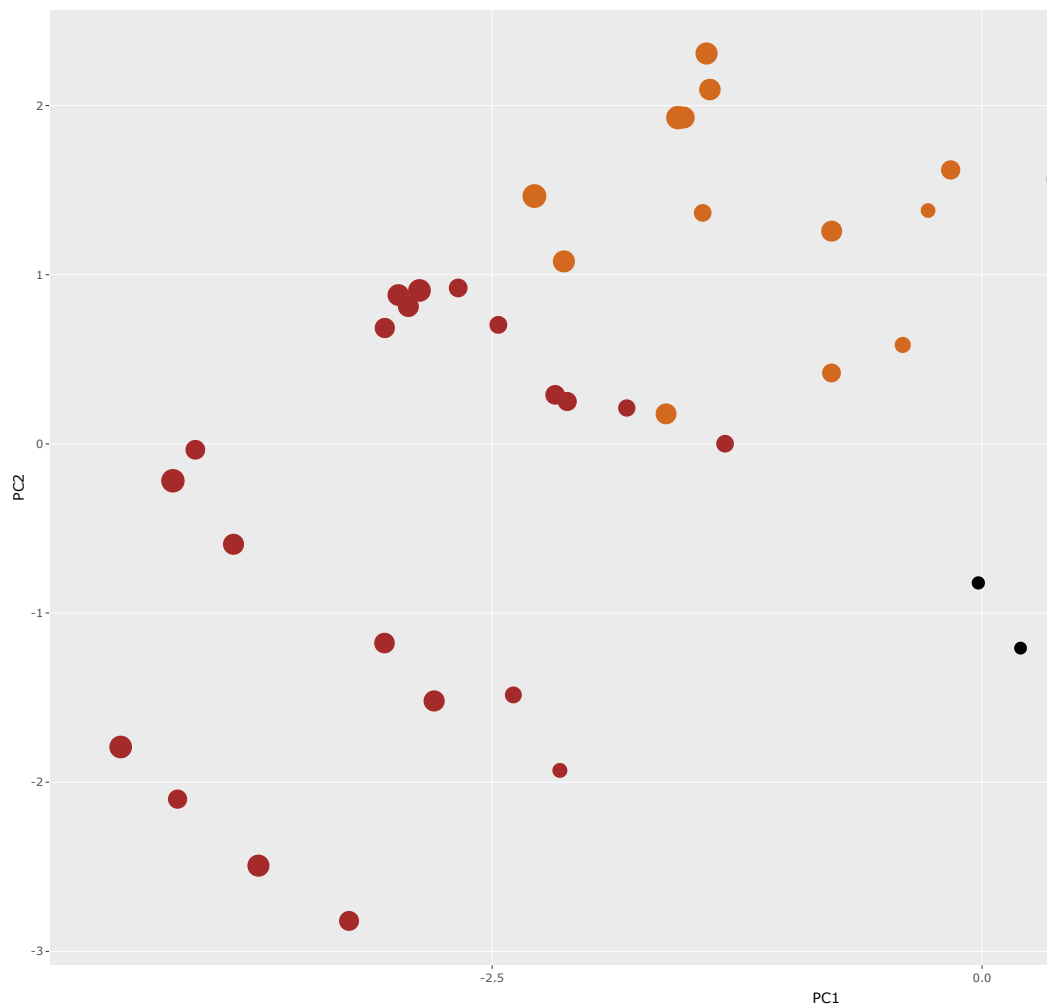
The following object is masked from 'package:stats':

`filter`

The following object is masked from 'package:graphics':

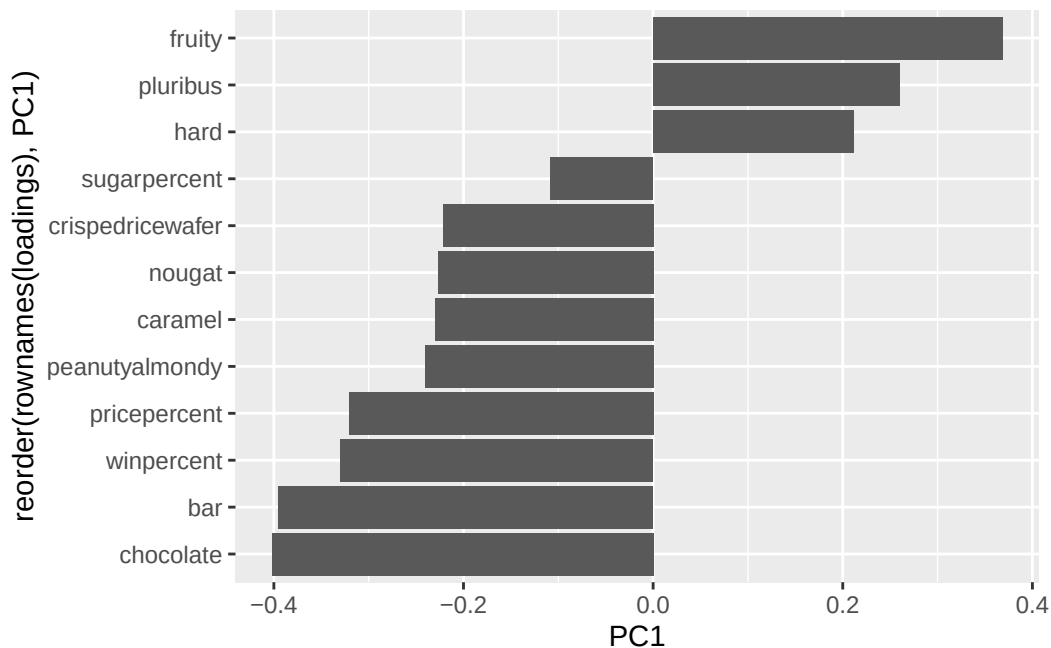
`layout`

`ggplotly(p)`



```
#Completed Q24 code
loadings <- as.data.frame(pca$rotation)

ggplot(loadings) +
  aes(x = PC1, y = reorder(rownames(loadings), PC1)) +
  geom_col()
```



SUMMARY

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy appears to contain chocolate, often in bar form and paired with additions, such as nuts, caramel, or nougat, rather than a simple fruity candy. The visual analysis showed that chocolate candies consistently ranked higher compared to fruity candies. The correlation analysis supports this finding that chocolate and fruity are strongly negatively correlated. The PCA integrates these patterns, with PC1 separating chocolate-based candies and fruity-based candies, which indicates that this is the main source of variation.